Multimodal Hateful Meme Detection: Mid-Project Report

Lexical Leaders
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https://github.com/MuditGupta2502/Meta-Hateful-Meme-Detection

Abstract

This mid-submission report presents our project direction and current progress on multimodal hateful meme detection. Our core idea is to detect harmful memes by jointly modeling image and text context with a frozen CLIP backbone and a trainable cross-attention fusion head. So far, we have completed the Phase 1 baseline and a stronger Phase 1 extension based on class-balanced sampling. On the balanced development split, the baseline reaches AUROC 0.7077 and the extension reaches AUROC 0.7306. We also completed a controlled ablation that validates cross-attention over late-fusion concatenation (+0.0399 AUROC). For the planned Phase 2, we will focus on deployment realism (OCR), selective encoder adaptation (LoRA/PEFT), and an ethical safety layer with human-in-the-loop intervention for uncertain or high-risk predictions.

1 Project Idea and Motivation

Hateful memes are difficult to detect because hateful intent usually emerges from *interaction* between image and text, not from one modality in isolation. A seemingly harmless image can become hateful with a specific caption, and a provocative sentence can be benign depending on visual context. This makes the task fundamentally multimodal.

Our project aims to build a practical and reproducible multimodal detection system that is both computationally efficient and extensible. Instead of full end-to-end fine-tuning of very large encoders, we freeze a strong pretrained backbone and train a smaller fusion/classification component. This design is suitable for academic constraints (limited GPU budget) while still allowing methodical experimentation.

Primary objective. Given meme image v and text t, learn a classifier

$$f(v, t) \to \hat{y} \in \{0, 1\},$$

where 1 denotes hateful and 0 denotes non-hateful.

Mid-project objectives.

- Build a stable multimodal baseline with frozen CLIP encoders and cross-attention fusion.
- Improve minority-class robustness under class imbalance.
- Validate architecture decisions using controlled ablations.
- Define a realistic second phase with safety and deployment-focused additions.

2 Background and Related Work

The Hateful Memes Challenge (Kiela et al., 2020) formalized this task with "benign confounders" so that unimodal shortcuts fail by design. This motivates fusion architectures that explicitly reason over image-text interactions.

Pretrained vision-language models, especially CLIP (Radford et al., 2021), provide strong aligned embeddings for image and text. However, direct zero-shot similarity scoring is typically insufficient for subtle hate detection because the decision boundary depends on dataset-specific context and label semantics.

Transformer attention (Vaswani et al., 2017) motivates our cross-attention fusion block, where each modality attends to the other before classification. For imbalanced classification, focal loss (Lin et al., 2017) provides hard-example emphasis that is particularly useful when minority errors are costly.

For the planned second phase, we consider parameter-efficient adaptation via LoRA (Hu et al., 2022) so that representation quality can improve without full encoder fine-tuning. We also include model-risk safeguards inspired by broader responsible-AI moderation practice (Weidinger et al., 2021).

2.1 Recent Multimodal Foundation-Model Context

Recent multimodal systems provide useful context for why we keep a parameter-efficient design at mid stage. BLIP-2 (Li et al., 2023) and Flamingo (Alayrac et al., 2022) show that frozen vision encoders with lightweight adapters can be highly competitive. Instruction-tuned vision-language assistants such as LLaVA (Liu et al., 2023) further indicate that alignment quality often depends more on adaptation strategy and data quality than on full end-to-end retraining.

This informs our project choices in two ways. First, we retain frozen CLIP features for compute efficiency and reproducibility in a classroom setting. Second, for planned Phase 2 we prioritize targeted adaptation (LoRA/PEFT + OCR realism + safety policy) over brute-force scaling.

2.2 Task-Specific Moderation Considerations

Unlike generic image-text understanding, hateful meme detection has asymmetric error costs: false negatives may allow harmful content to pass, while false positives may suppress benign speech. Because of this asymmetry, we treat moderation as a *risk-sensitive decision process* rather than pure classification. This is why our plan combines predictive metrics (AUROC/F1) with policy-aware operational metrics (escalation rate, reviewer agreement, and auditability).

3 Dataset

We use the Meta AI Hateful Memes dataset (Kiela et al., 2020). The dataset provides image paths, meme text, and binary labels for train/dev splits; the public test split is unlabeled.

Table 1: Dataset split statistics used in this project.

Split	Total	Hateful	Non-hateful	Ratio
Train	8,500	3,050	5,450	1.79:1
Dev	500	250	250	1:1
Test	1,000	–	–	–

Observed properties from EDA.

- Meme text is short (mean ≈ 11.7 words), with long-tail stylistic noise.
- The train split is moderately imbalanced, requiring explicit mitigation.
- The development split is balanced and therefore suitable for robust threshold-free ranking metrics (AUROC).

3.1 Exploratory Data Analysis Summary

We performed an initial EDA pass to verify whether training assumptions match the actual data distribution.

Table 2: Word-count statistics on the training split.

Class	Mean	Median	Max	95th pct.
Non-hateful	11.2	10	69	23
Hateful	12.8	11	70	26
Overall	11.7	10	70	24

These numbers confirm that meme text length is comfortably inside CLIP token limits, so truncation is not a dominant error source.

Table 3: Top frequent content tokens (stop-words removed).

Hateful cluster examples	Non-hateful cluster examples
"white", "black", "muslim"	"love", "happy", "friend"
"illegal", "parasite", "send"	"birthday", "dog", "cat"
"terror", "country", "back"	"food", "work", "today"

The lexical separation is non-trivial but incomplete, reinforcing the need for multimodal context instead of text-only filtering.

4 Proposed Methodology and Architecture

4.1 Model Overview

Our baseline architecture has three stages:

1. **Frozen multimodal encoding:** CLIP ViT-B/32 encodes image and text into 512-d vectors.
2. **Bidirectional cross-attention fusion:** image queries attend to text features and text queries attend to image features.
3. **Classification head:** a lightweight MLP predicts hateful vs non-hateful.

Only the fusion and classifier layers are trainable (about 2.4M parameters), while the CLIP backbone remains frozen (about 151M parameters).

4.2 Training Design

Loss. Weighted focal loss with class weights $[1.0,\ 1.786]$ and $\gamma = 2$.

Optimizer and schedule. AdamW, learning rate 10^{-4}, cosine decay with linear warmup.

Class imbalance strategy.

- Baseline: weighted focal loss only.

- Phase 1 extension: weighted focal loss + `WeightedRandomSampler` to create near-balanced sampling per epoch.

4.3 Ablation Methodology

To test whether cross-attention itself is useful, we compare:

- **Late Fusion:** concatenation of frozen CLIP image/text embeddings + MLP.
- **Cross-Attention Fusion:** full proposed architecture.

All other training choices are held fixed.

4.4 Implementation Footprint

Table 4: Parameter and compute footprint (current implementation).

Component	Parameters	Trainable
CLIP ViT-B/32 backbone	∼151.3M	No
Cross-attention fusion block	∼2.1M	Yes
Classification head (MLP)	∼0.3M	Yes
Total	∼153.7M	∼2.4M (1.6%)

This parameter-efficiency is important for reproducibility on limited hardware (Kaggle T4 class GPUs) while still allowing architecture-level experimentation.

4.5 Core Objective Functions

The classifier is trained with weighted focal loss:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t),$$

where p_t is the model probability assigned to the true class, $\gamma = 2$, and α_t follows class imbalance.

For a batch $\mathcal{B}$, the optimization objective is:

$$\min_{\theta} \frac{1}{|\mathcal{B}|} \sum_{(v_i, t_i, y_i) \in \mathcal{B}} \mathcal{L}_{\text{focal}}\big(y_i, f_\theta(v_i, t_i)\big).$$

In the cross-attention module, image token representations V attend to text tokens T, and text attends back to image:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

This bidirectional interaction helps the model capture meaning that only emerges from image-text pairing.

5 Evaluation Options and Metrics

We evaluate on the 500-sample balanced dev split.

Primary metric: AUROC (threshold-independent ranking quality).

Secondary metrics: Accuracy, F1 (hateful class), Macro-F1, confusion matrix.

Why these metrics.

- AUROC is robust to threshold choice and is widely used for this task setup.
- Macro-F1 captures class-balance quality and guards against one-class dominance.
- Confusion matrices support practical moderation analysis (false positives vs false negatives).

We report best checkpoints by AUROC and track metric trajectories across epochs for overfitting diagnosis.

5.1 Extended Evaluation Protocol (Final Phase Plan)

Beyond raw point estimates, final-phase evaluation will include protocol-level checks so that reported gains are robust and operationally meaningful.

Threshold calibration options.

- Fixed threshold (0.5) for reproducible baseline reporting.
- Dev-optimized threshold maximizing Macro-F1 under class-balance constraints.
- Two-threshold policy for abstention: auto-accept below τ_{low}, auto-reject above τ_{high}, and human escalation in between.

Uncertainty-aware reporting. To avoid over-interpreting small differences, we will report bootstrap confidence intervals for AUROC and Macro-F1 on the dev set:

$$\text{CI}_{95\%}(m) = [q_{0.025}(m^*),\ q_{0.975}(m^*)],$$

where m^* denotes metric values across bootstrap resamples.

Stress-test slices. We will compute metrics on targeted slices (OCR-noisy text, sarcasm-heavy samples, and identity-term-heavy samples) to identify whether aggregate gains hide subgroup regressions.

Table 5: Planned final-phase reporting bundle.

Artifact	Purpose
Overall dev metrics table	Primary model ranking and core comparability
Threshold-policy table	Operational tradeoff (automation vs escalation)
Bootstrap CI table	Statistical stability of reported gains
Slice-wise error table	Detect hidden regressions on difficult subsets
Reviewer decision log summary	Validate human-in-the-loop practicality

6 Interim Experimental Results

6.1 Phase 1 Baseline (10 epochs)

6.2 Phase 1 Extension (20 epochs, balanced sampler)

6.3 Interim Findings

- **Consistent gain from balanced sampling:** the extension improves AUROC by +0.0229 over baseline.
- **Best operating point appears early:** AUROC peaks around epoch 8 in the extension, after which validation loss rises sharply.
- **Ablation confirms fusion design:** explicit cross-attention outperforms simple concatenation by +0.0399 AUROC.

6.4 Qualitative Error Analysis (Interim)

We manually inspected representative false positives and false negatives from dev predictions and grouped them into recurring error modes.

These observations guide planned Phase 2 work: OCR robustness, stronger context adaptation (LoRA), and abstention plus human review for uncertain cases.

7 Progress So Far

7.1 Completed Work

Interpretation. The project has passed key mid-stage milestones: (i) stable multimodal baseline, (ii) measurable gain from class-balanced training, and (iii) architectural validation through ablation. The next bottleneck is not training stability but deployment realism and risk-sensitive moderation quality.

7.2 Ongoing Work

Current work focuses on three tracks:

- **Failure taxonomy:** identify recurring failure modes (sarcasm, OCR-like text distortions, rare cultural context).
- **Threshold policy design:** define operating thresholds for automatic accept/reject and *escalate-to-human* cases.
- **Implementation cleanup:** improve experiment tracking and reproducibility notes for final submission.

7.3 Planned Phase 2: Counterfactual Editing + Safety + Realism

Planned Phase 2 expands the current system with practical moderation safeguards and generation-based counterfactual intervention.

Human-in-the-loop intervention. For uncertain or high-risk cases, the model will abstain and escalate to manual review. We will define:

- confidence bands for auto-decision vs human escalation,
- reviewer guideline checklist,
- audit logs for traceability and post-hoc correction,
- feedback loop to refine thresholds and retraining datasets.

Additional deliverables.

- Counterfactual text editing module to minimally rewrite hateful meme text into non-hateful alternatives while preserving core semantics.
- Counterfactual image generation module to produce non-hateful image variants from hateful memes via CLIP latent editing and decoding.
- OCR pipeline for end-to-end inference.
- LoRA-based adaptation of selected CLIP blocks.
- Robustness tests with noisy overlays and spelling perturbations.

Counterfactual image-generation direction. Given mentor feedback, we will prototype an embedding-space intervention pipeline: encode meme image to CLIP representation, identify and suppress hate-associated latent directions (including SAE-style sparse feature editing as a candidate), then decode the edited representation into a non-hateful image candidate. The edited image will be evaluated for (i) label flip away from hateful, (ii) semantic preservation, and (iii) visual quality.

Table 6: Main hyperparameters for the completed mid-phase experiments.

Category	Setting
Backbone	`openai/clip-vit-base-patch32` (frozen)
Fusion module	Bidirectional cross-attention, 4 heads
Classifier head	3-layer MLP with dropout
Loss	Weighted focal loss, $\gamma = 2$, class weights $[1.0, 1.786]$
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Scheduler	Cosine decay with linear warmup
Batch size	32
Phase 1 baseline	10 epochs, warmup 100
Phase 1 extension	20 epochs, warmup 200, weighted random sampler
Primary checkpoint criterion	Validation AUROC
Runtime footprint	Approximately 35 minutes (10-epoch run, Kaggle T4)

Table 7: Phase 1 baseline: epoch-wise validation trajectory.

Epoch	Train Loss	Val Loss	Val AUROC	Val F1	Macro-F1
1	0.298	0.356	0.639	0.600	0.531
2	0.266	0.346	0.677	0.644	0.611
3	0.245	0.344	0.681	0.665	0.633
4	0.229	0.354	0.695	0.650	0.629
5	0.219	0.358	0.702	0.695	0.664
6	0.207	0.359	0.706	0.689	0.670
7	0.196	0.367	0.707	0.682	0.666
8	0.186	0.381	0.703	0.673	0.664
9	0.178	0.389	0.708	0.664	0.658
10	0.170	0.395	0.705	0.655	0.653

7.4 Planned Phase 2 Experiment Matrix

To keep Phase 2 systematic, we predefine an experiment matrix instead of ad-hoc runs.

8 Ethical AI and Safety Plan

Because this project targets harmful content moderation, we explicitly include a safety plan in technical development.

8.1 Human Intervention Policy

- **Low-confidence abstention:** if model confidence falls into a predefined uncertainty band, no automatic moderation action is taken.
- **Manual escalation:** uncertain/high-impact samples are routed to human reviewers.
- **Dual-review for critical cases:** content involving protected attributes can trigger secondary review.
- **Auditability:** model score, threshold band, and reviewer decision are logged for later audit.

8.2 Safety Evaluation Options

In planned Phase 2, we will add safety-specific evaluation dimensions alongside AUROC/F1:

- **Escalation rate:** fraction of samples routed to human review.
- **Reviewer agreement:** consistency between independent reviewers for escalated cases.
- **Harm-weighted error analysis:** separate accounting of false negatives in high-risk categories.
- **Turnaround viability:** average time from escalation to final decision.

This framing ensures the project remains not only accurate but also safer for real moderation workflows.

8.3 Risk Register and Mitigation

9 Success Criteria and Final Evaluation Plan

To avoid ambiguous project completion criteria, we define quantitative and process-level targets in advance.

9.1 Quantitative Targets

- **Primary ranking quality target:** improve or maintain AUROC relative to the current Phase 1

Table 8: Phase 1 extension: epoch-wise validation trajectory (best AUROC at epoch 8).

Epoch	Train Loss	Val Loss	Val AUROC	Val F1	Macro-F1
1	0.301	0.314	0.6696	0.6918	0.5441
2	0.251	0.318	0.6831	0.6875	0.5660
3	0.229	0.319	0.7005	0.7018	0.6206
4	0.208	0.342	0.7049	0.6985	0.6349
5	0.188	0.345	0.7206	0.7018	0.6532
6	0.164	0.408	0.7147	0.6991	0.6619
7	0.135	0.555	0.7251	0.6534	0.6520
8	0.113	0.538	0.7306	0.6667	0.6700
9	0.091	0.695	0.7286	0.6345	0.6512
10	0.077	0.736	0.7301	0.6126	0.6516
11	0.062	0.820	0.7294	0.6000	0.6429
12	0.053	0.890	0.7305	0.6233	0.6600
13	0.047	0.986	0.7221	0.5701	0.6242
14	0.039	0.953	0.7286	0.6154	0.6554
15	0.036	1.020	0.7247	0.5945	0.6418
16	0.029	1.161	0.7258	0.5841	0.6365
17	0.028	1.158	0.7268	0.5874	0.6387
18	0.029	1.178	0.7274	0.5855	0.6383
19	0.025	1.184	0.7269	0.5808	0.6342
20	0.028	1.182	0.7271	0.5794	0.6324

Table 9: Interim failure taxonomy from manual inspection.

Category	Typical pattern
Sarcasm / irony ambiguity	Surface text appears harmless but implied intent is hateful in cultural context.
Context beyond meme frame	External political/cultural references not visible in image-text pair.
Text distortion / stylization	Deliberate misspellings, split words, and symbol substitutions.
Entity confusion	Model latches on identity words without full relational context.
Image cue over-reliance	Salient visual feature dominates even when text negates hateful intent.

extension benchmark (0.7306) after introducing OCR and safety policies.

- **Balanced performance target:** maintain Macro-F1 close to AUROC improvements so that gains are not caused by one-class bias.
- **Safety-operational target:** ensure uncertain/high-risk routing does not collapse coverage (i.e., escalation rate remains practical for human moderation bandwidth).

9.2 Operational and Reporting Targets

- **Reproducibility target:** every table in the final report must map to a runnable command and a saved configuration.
- **Traceability target:** reviewer interventions for escalated cases should be logged with reasons and outcome labels.
- **Error-analysis target:** final write-up should include at least one validated mitigation for each major failure category in Table 9.

10 Team Organization and Execution Workflow

We split work by technical tracks while keeping weekly integration points to avoid drift between modeling, data processing, and safety objectives.

Execution protocol.

- Weekly experiment freeze: only validated runs enter report tables.
- Shared artifact naming convention: run ID + config hash + timestamp.
- Cross-track review before merge: modeling updates and safety policy updates are reviewed together to avoid metric-only optimization.

This workflow is designed to keep the final submission balanced across model quality, safety, and reproducibility instead of optimizing a single headline metric.

11 Tentative Timeline

12 Repository and Reproducibility

The public repository is:

Table 10: Progress summary (completed, ongoing, and planned).

Work Item	Status	Notes
Dataset loading and validation	Completed	Verified paths, labels, and train/dev split consistency.
Baseline model (frozen CLIP + cross-attention)	Completed	End-to-end training and evaluation pipeline finalized.
Baseline experiment (10 epochs)	Completed	Best AUROC 0.7077; best Accuracy 67.0%.
Phase 1 extension (balanced sampler, 20 epochs)	Completed	Best AUROC 0.7306 at epoch 8; +0.0229 over baseline.
Ablation (late fusion vs cross-attention)	Completed	Cross-attention +0.0399 AUROC over late fusion.
Error analysis and case categorization	Ongoing	Collecting examples for text-dominant vs image-dominant failures.
Phase 2 safety design (human intervention policy)	Ongoing	Defining confidence thresholds and escalation flow.
OCR-based end-to-end inference	Planned	Replace dataset text dependence with OCR extraction at inference.
LoRA/PEFT adaptation experiments	Planned	Adapt selective CLIP layers without full fine-tuning.
Safety evaluation protocol	Planned	Measure abstention, escalation rate, and reviewer agreement.

Table 11: Internal performance comparison on dev split.

Model Variant	AUROC	Accuracy
Random baseline	0.5000	50.0%
CLIP zero-shot (cosine)	≈0.52	≈52%
Phase 1 baseline	0.7077	67.0%
Phase 1 extension	0.7306	67.0%

Table 12: Ablation result (same training recipe, 10 epochs).

Fusion Strategy	AUROC	Macro-F1
Late Fusion	0.6682	0.6127
Cross-Attention Fusion	0.7081	0.6320
Δ (CrossAttn - LateFusion)	+0.0399	+0.0193

https://github.com/MuditGupta2502/Meta-Hateful-Meme-Detection

Main reproducibility commands used in this mid phase are:

- `python train.py -epochs 10 -batch_size 32 -lr 1e-4`
- `python train.py -config configs/config_smote.yaml`
- `python train_ablation.py -config configs/config.yaml -epochs 10`
- `python train_ablation.py -variants late_fusion cross_attention`

13 Conclusion

This mid-submission confirms that our project direction is viable and evidence-driven. We have already completed the core multimodal architecture and demonstrated consistent gains through a Phase 1 extension and ablation experiments. The planned Phase 2 is now focused on what the current system still lacks for realistic deployment: OCR integration, parameter-efficient adaptation, and explicit ethical safety mechanisms with human intervention on uncertain/high-risk samples.

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Table 13: Planned Phase 2 experiment matrix (design before execution).

Track	Experiment ID	Core change	Evaluation focus
OCR realism	P2-OCR-1	OCR extraction with default parser	End-to-end AUROC vs text-field oracle
OCR realism	P2-OCR-2	OCR + normalization (case, punctuation, spacing)	Robustness under noisy text rendering
Text counterfactuals	P2-CF-TEXT-1	Minimal hateful-to-benign text rewrite generator	Label flip rate and semantic preservation
Image counterfactuals	P2-CF-IMG-1	CLIP embedding intervention + decoder generation	Non-hateful conversion rate and visual fidelity
Image counterfactuals	P2-CF-IMG-2	SAE-style sparse latent editing before decode	Edit controllability and artifact rate
PEFT adaptation	P2-LORA-1	LoRA on final CLIP attention blocks	AUROC/F1 gain per trainable parameter
PEFT adaptation	P2-LORA-2	LoRA rank sweep (r=4,8,16)	Cost-quality tradeoff
Safety policy	P2-SAFE-1	Confidence band with abstention	Escalation rate and retained accuracy
Safety policy	P2-SAFE-2	High-risk keyword trigger for human review	Harm-weighted error reduction
Human feedback	P2-HITL-1	Reviewer decision logging	Agreement and correction usefulness
Stress testing	P2-ROB-1	Character noise and overlay perturbations	Performance degradation profile

Table 14: Current project risks and mitigations.

Risk	Mitigation
Overfitting after peak epoch	Early-stop checkpoints by AUROC and restrict deployment to validated windows.
OCR extraction noise	Add text normalization and evaluate noisy synthetic overlays before rollout.
Counterfactual image artifacts	Add visual-quality filtering and human safety review before accepting generated edits.
Unsafe false negatives	Introduce abstention and mandatory manual review for uncertainty/high-risk bands.
Compute budget limits	Use PEFT (LoRA) instead of full backbone fine-tuning.
Annotation subjectivity	Reviewer guidelines + disagreement resolution + audit logs.

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Table 15: Final evaluation checklist aligned with project objectives.

Objective	Acceptance signal	Evidence artifact
Strong multimodal baseline	Stable re-run with near-identical trend	Training logs + checkpoint metrics
Class-imbalance robustness	Extension outperforms baseline on AUROC/Macro-F1	Comparative metrics table and script output
Architectural validation	Cross-attention keeps advantage over late fusion	Ablation result table and run logs
Deployment realism	OCR-enabled path evaluated end-to-end	OCR experiment report and error examples
Counterfactual intervention quality	Text and image counterfactual pipelines show controlled hateful-to-benign edits	Counterfactual examples, metrics, and failure cases
Ethical safety integration	Human-in-the-loop escalation policy operational	Escalation statistics + reviewer protocol
Project reproducibility	Commands and configs recreate all major results	README + config snapshots + command list

Table 16: Working allocation for the remaining project phase.

Track owner(s)	Responsibility
Modeling track	LoRA/PEFT experiments, training stability checks, ablation updates, and text counterfactual generation.
Data/OCR track	OCR pipeline integration, normalization, noisy text stress testing, and image counterfactual decoding pipeline.
Safety track	Confidence thresholds, escalation logic, reviewer guideline sheet, audit logs.
Evaluation/report track	Metrics consolidation, error analysis write-up, timeline tracking.

Table 17: Tentative timeline from mid-submission to final evaluation.

Time Window	Planned Deliverables
Mar 9–Mar 16	Finalize failure taxonomy; freeze Phase 2 safety policy draft; define counterfactual text/image metrics.
Mar 17–Mar 24	Implement OCR pipeline, confidence-based escalation logic, and text counterfactual baseline.
Mar 25–Apr 2	Prototype image counterfactual pipeline (CLIP latent editing + decoder, including SAE-style variant).
Apr 3–Apr 10	Run LoRA/PEFT experiments and integrated evaluation of counterfactual, OCR, and safety modules.
Apr 11–Apr 18	Final safety review of generated counterfactuals, consolidate results, finalize report/figures, prepare viva notes.